AIA Brochure Q&A: a grounded RAG system with an agent

LLM course capstone · code: github.com/Calvinkub/aia-insurance-rag · chat model `gemma-4-E4B-it`, embeddings `bge-m3` via the class proxy

1. Problem and audience

Insurance brochures hold the facts a customer needs (waiting periods, issue ages, benefit percentages, fees), but those facts sit in tables and disclaimers spread across many PDFs. A general-purpose chatbot answers from memory, which for a specific Thai product means a confident guess. This project answers questions about AIA Thailand products *only from the brochure text*, cites the page, and says "the brochures do not say this" when the evidence is absent.

Who it is for: a prospective policyholder or an advisor who wants a fast, checkable answer, not a recommendation. It is a course project, not insurance advice. The brochures are dated 2019–2022 and may be out of date.

Why a reference app cannot do this: the corpus is 16 public AIA Thailand PDFs, downloaded by `scripts/fetch_corpus.py`. Fifteen yield usable English text (94 pages, 290 chunks); the closed-book model has no reliable knowledge of them, and the evaluation below shows it (7% accuracy when forced to guess).

2. Design

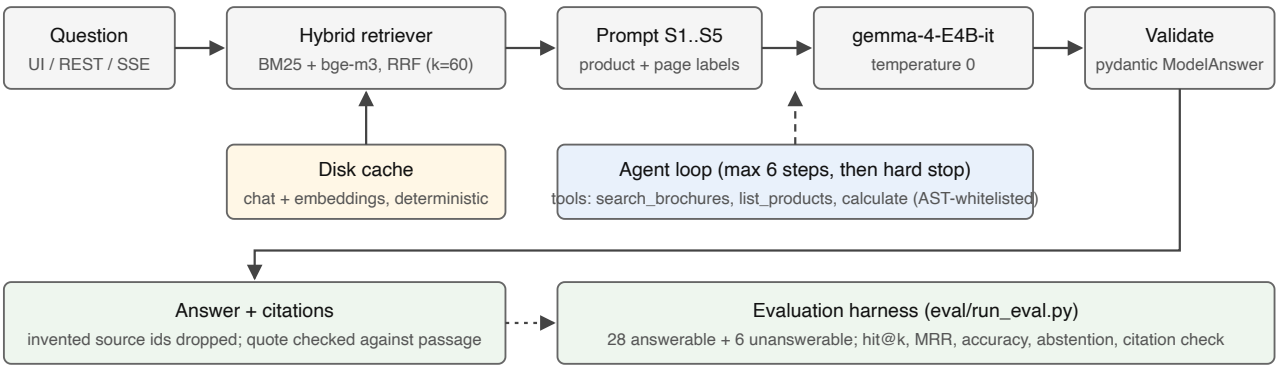


Figure 1. Request path (top), the agent as an alternative loop over the same retriever (blue), and the evaluation harness that scores the output of the path.

Course technique	How it is used here	Why
Prompting + structured output with schema validation	The model must return JSON matching `ModelAnswer` (answerable, answer, citations). Code fences are stripped, one repair retry runs, otherwise the system abstains.	Downstream code never trusts free text.
RAG with citations	Sources are numbered S1..S5. The model cites ids, not page numbers; code resolves ids to product and page, drops invented ids, and marks a citation `verified` only if its quote occurs in the cited passage (alphanumeric-normalised).	A citation the user cannot check is decoration.
Retrieval and answer evaluation	Hand-labelled gold set, page-level hit@k and MRR, answer accuracy, abstention, citation checks, three baselines.	Required family; also how every design choice below was made.
Agent loop with tools	Native function calling with an allow-list, pydantic-validated arguments, a 6-step cap, and tool output treated as untrusted. `calculate` evaluates arithmetic through an AST whitelist (no names, calls or attributes; exponent capped).	Lets multi-step questions (compare two products, compute a benefit) use search plus arithmetic safely.
Served interface	FastAPI: `POST /ask`, SSE `GET /ask/stream` and `/agent/stream`, minimal UI at `/`.	Streaming shows retrieved sources first and tool calls as they happen.

Hybrid retrieval was chosen over dense-only because brochure questions mix exact tokens (numbers, product names) with paraphrase; reciprocal-rank fusion needs no score calibration. The API key and base URL come only from the environment, and no secret is committed.

3. Evaluation

Test set (`eval/gold.jsonl`): 28 answerable questions, each with a verbatim evidence quote, gold product and page, and required keywords; plus 6 unanswerable questions that are topically close but absent from the corpus (for example Vitality reward tiers). `eval/check_gold.py` verifies every evidence quote against the extracted page text and checks that the unanswerable terms do not occur in the corpus. Questions were drafted with AI assistance from the brochure text; the quote check keeps the labels honest, but they are not independent human labels.

Metrics. Retrieval: hit@k and MRR, where a hit is a top-k chunk from the gold product on a gold page. Answers: *accuracy* (all gold keywords appear as whole tokens), *false abstain* (answerable question declined), *abstain on unanswerable*, *cites gold page*, and *citation verified* (share of citations whose quote really occurs in the passage). All runs use temperature 0 with an on-disk cache so results reproduce.

Baselines. (a) closed-book, may abstain; (b) closed-book, forced to guess; (c) BM25 RAG with the same chunks. The first scores 0 only because the model refuses everything, so (b) is the informative closed-book baseline.

4. Results

4.1 End-to-end answers (chunks 180 words / 40 overlap, k = 5)

system	accuracy	false abstain	abstain on unanswerable	cites gold page	citation verified
closed-book, may abstain (baseline)	0.00	1.00	1.00	–	–
closed-book, forced guess (baseline)	0.07	0.29	0.50	–	–
BM25 RAG, no floor (baseline)	0.93	0.04	1.00	0.93	0.95
hybrid RAG, no floor (shipped)	**0.93**	**0.00**	**1.00**	**0.93**	**1.00**
hybrid RAG, `min_score` = 0.65	0.89	0.04	1.00	0.89	1.00

Table 1. n = 28 answerable, 6 unanswerable. One question = 0.036 accuracy.

- **Retrieval is what makes it work.** Forced-guess closed-book gets 2 of 28 right and answers 3 of 6 unanswerable questions anyway. With retrieval: 26 of 28, and no answer to any unanswerable question.
- **Measured improvement over the naive default** (BM25, same chunks): hybrid raises MRR 0.832 to 0.882, hit@1 0.79 to 0.82, citation-verified rate 0.95 to 1.00, and removes the one false abstain. With n = 28 this is one or two questions: a direction, not a significant result.
- `min_score` **was a negative result.** A threshold of 0.65 looked free in a query-level sweep (kept all 28 answerable questions, blocked 4 of 6 unanswerable ones). Applied as a per-chunk floor it removed a correct chunk (cosine 0.62) and cost one answer (0.93 to 0.89). The model already declines all 6 unanswerable questions, so the default is 0.

4.2 Retrieval sweep (n = 28, k = 5)

chunk words/overlap	mode	hit@1	hit@3	hit@5	MRR
80/20	bm25 / dense / hybrid	0.75 / 0.75 / 0.79	0.93 / 0.93 / 0.93	0.93 / 0.96 / 0.93	0.839 / 0.842 / 0.857
180/40 (chosen)	**bm25 / dense / hybrid**	**0.79 / 0.79 / 0.82**	**0.86 / 0.93 / 0.93**	**0.93 / 0.96 / 0.96**	**0.832 / 0.852 / 0.882**
350/60	bm25 / dense / hybrid	0.75 / 0.68 / 0.75	0.96 / 0.86 / 0.96	0.96 / 0.96 / 0.96	0.845 / 0.787 / 0.857

Table 2. Chunk size and retrieval mode. 350-word chunks hurt dense retrieval most.

4.3 Chunking experiments

The baseline E0 cuts fixed 180-word windows: 74% of chunks end mid-sentence and about 15% of the text is disclaimers repeated on every page. Four techniques came from a NotebookLM literature pass (web sources; figures not independently verified) and were each measured on the same gold set. Late chunking was skipped because it needs a different embedding model than the proxy offers; semantic chunking was skipped because it reportedly fragments short text.

	pipeline	chunks	hybrid MRR	dense MRR	answer acc.	citations verified
E0	**baseline: 180-word windows**	**290**	**0.882**	**0.852**	**0.93**	**1.00**
E1	+ drop repeated boilerplate, sentence-aware packing	249	0.893	0.878	0.89	0.97
E2	PyMuPDF4LLM markdown tables, header repeated per chunk	262	0.875	0.788	0.96	0.92
E3	E2 + LLM-written context sentence per chunk	262	0.873	0.848	0.96	0.88
E4	parent-child: ~60-word children, whole page returned	650	0.889	0.923	0.93	0.95

Table 3. Answer accuracy and citations verified are for hybrid retrieval.

No variant clearly beats the baseline. Accuracy moves by ±1 question (25–27 of 28) and MRR by 0.01–0.04, all inside noise for n = 28. E1 cuts mid-sentence endings from 74% to 13% of chunks but did not raise accuracy. E2 and E3 fixed one question (A24) but lowered the share of verifiable citations from 1.00 to 0.88–0.92; the cause was not investigated. E4 with dense-only retrieval was the best retriever (hit@3 = hit@5 = 1.00, MRR 0.923), but its answers were not evaluated, to avoid tuning further on the same questions, so this is a hypothesis for a fresh question set, not a result. E0 stays the default; the variants are opt-in.

5. Failure cases

id	question	what happened	cause
A07	How many severe-stage critical illnesses does AIA CI Care cover?	Answered 44; the gold label says 40.	The brochure contradicts itself: page 1 says "Cover 44 severe stage critical illnesses", the page-2 table header says "Severe Stage (40 Diseases / Treatment)". The gold label follows the table; the model followed page 1, which is a supported reading. This is a source conflict, not a hallucination. The ideal behaviour is to flag the conflict, and the system does not.
A24	Which premium payment periods can be chosen for Infinite Wealth Prestige?	Answered with top-up payment *frequency* (monthly, quarterly).	The gold chunk (page 3) ranked 5th; the model answered from a higher-ranked passage that uses a different sense of "payment period".
A17 (floor only)	Maximum income tax deduction for Smart Select?	False abstain.	The `min_score` = 0.65 floor dropped the evidence chunk.

> **Correction on record.** I first diagnosed A07 as the model inventing "44" because a flattened table lost its structure. Re-reading page 1 showed that the number is in the brochure. The diagnosis, and the README that repeated it, were wrong and have been corrected. The lesson for the evaluation: a label can be the contested side, so each failure needs a look at the source before blaming the model.

6. Limitations and next steps

- **Small, self-built eval, tuned on itself.** Chunk size, retrieval mode and `min_score` were chosen by looking at the same 34 questions. There is no held-out set, so the shipped numbers are optimistic, and one-question differences are noise.
- **Keyword scoring is a proxy.** It can credit the right number in the wrong context and miss a correct paraphrase. One gold keyword was a word stem that wrongly failed a correct answer; it was changed to a full word. No LLM judge was used.

- **Conflicting sources are not flagged** (A07), and PDF tables flatten with the default parser. The table parser (E2) did not measurably help here, so the effect of flattening is untested rather than disproved.
- **English brochures only.** Three files labelled English are actually Thai and extract as garbage, so they are filtered out; some products (term life, endowment, pay-life) are thin or missing.
- **The agent is not quantitatively evaluated.** It was exercised by hand (multi-step comparison, argument-rejection tests); in one manual run it multiplied by hand instead of calling `calculate`.
- **One model family does everything**, with no independent judge, and depends on the class proxy. The repo was built in one working session, so its history is a short run of logical commits.

Next steps: detect and report conflicting figures across retrieved passages; test dense-only parent-child retrieval on a fresh question set; find out why table-based variants lower citation verification; write a larger held-out gold set that the system was not tuned on; add an agent gold set; support Thai.